```
import cv2

# Read uploaded video
cap = cv2.VideoCapture('lab video.mp4')

# Video properties
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)

# Output videos
fourcc = cv2.VideoWriter_fourcc(*'mp4v')

slow_out = cv2.VideoWriter(
    'slow_motion.mp4',
    fourcc,
    fps / 2,
    (width, height)
)

fast_out = cv2.VideoWriter(
    'fast_motion.mp4',
    fourcc,
    fps * 2,
    (width, height)
)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    slow_out.write(frame)
    fast_out.write(frame)
```

```
cap.release()
slow_out.release()
fast_out.release()

print("Slow motion video saved as slow_motion.mp4")
print("Fast motion video saved as fast_motion.mp4")
```

```
Slow motion video saved as slow_motion.mp4
Fast motion video saved as fast_motion.mp4
```